Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 1 of 11 · Question 5 part b.i · 4 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
5	(a)	*hepatic vein* vena cava; pulmonary artery; pulmonary vein; aorta; renal artery;	Credit correct order	5
	(b) (i)	Any four from: • as time spent exercising increases, pulse rate increases; • increase flow of blood (to muscles/cells); • more oxygen/glucose/energy; • more (aerobic) respiration; • more carbon dioxide removed/reference to reduced build-up of lactic acid;	Allow more blood pumped Allow less anaerobic respiration	4
	(ii)	Any six from • take pulse before exercise/resting pulse; • finger at wrist/neck; • carry out exercise; • count/record pulse (after exercise); • wait for pulse rate to drop back to normal; • repeat exercise; • reference to control variables e.g. same person/gender/exercise/intensity of exercise/weather conditions;	Allow reference to use of electronic device to measure pulse rate Allow adequate time frame e.g. 5 minutes Allow reference to reliability e.g. more people	6

Total 15

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 1 of 11 · Question 5 part b.i · 4 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 2 of 11 · Question 8 part c · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

8(a)	<table><tr><th rowspan="2">Organ</th><th rowspan="2">Mass in kg</th><th colspan="2">Rate of blood flow in cm³ per minute</th></tr><tr><th>through the whole organ</th><th>per 100g of the organ</th></tr><tr><td>kidneys</td><td>0.3</td><td>1200</td><td>400</td></tr><tr><td>skeletal muscles</td><td>25.0</td><td>750</td><td>3</td></tr><tr><td>liver</td><td>1.5</td><td>1500</td><td>100</td></tr></table> kidney 1200÷3; 400; skeletal muscle 250 × 3; 750;	correct answer award full marks Ecf for each calculation allow correct answer to wrong figures	**(4)**

Question Number	Answer	Additional guidance	Mark
8(b)	Any four from: • (receives blood from) hepatic artery; • (receives blood from) hepatic portal vein; • homeostasis; • reference to blood sugar regulation; • detoxification (of alcohol)/urea formation/deamination;	For mp 1 and 2 allow receives blood from two vessels for one mark	**(4)**

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 2 of 11 · Question 8 part c · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
8(c)	Any four from: • less blood flow to kidney/liver; • increased blood flow to skeletal muscles; • (muscles have) greater energy demand/more energy needed/to reduce oxygen debt/lactic acid; • (so) more oxygen/glucose needed; • for (aerobic) respiration;		**(4)**

Total 12 Marks

Paper Total 90 Marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 3 of 11 · Question 9 part a.ii · 4 marks

2020 JAN · Paper 01 · find part a.ii in this mark scheme extract

number				
9 (a)	(i)	gives competitors an unfair advantage/form of cheating (1)		1
	(ii)	Any four from • increases heart rate (1) • for greater blood flow/increases blood pressure (1) • glycogen to glucose (in muscles/liver)(1) • causes dilation of air passages/increased rate/depth of breathing for increased oxygen uptake (1) • redirects blood flow to muscles/brain/heart/lungs (1) • increases aerobic respiration (1)		4
(b)	(i)	• beta-blocker attaches to/binds with (adrenaline/norepinephrine/epinephrine) receptor sites (1) • binding of adrenaline/norepinephrine / epinephrine inhibited/prevented/reduced (1) • reduced response in cardiac muscle/tissue cells/named reduced response e.g. lower heart rate (1)	ignore ref to active site	3
	(ii)	Any two from: • slower heart beat (1) • reduced blood flow/oxygen/glucose to skeletal muscles (1) • slower rate of aerobic respiration (1) • less energy (for activity) (1)		2

Total for Question 9 = 10 marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 4 of 11 · Question 2 part d · 1 mark

2020 JAN · Paper 02 · find part d in this mark scheme extract

Number			
2 (a) (i)	column headings(1) units in column headings(1) values correctly written in columns(1) Type of activity / exercise: at rest; walking; jogging; running Pulse/heart rate in beats per minute: 78; 95; 138; 146	Allow bpm for beats per minute	3
(b)	Any four from: • take pulse rate at rest(1) • using monitor/from wrist/neck(1) • walk (for one minute)(1) • take pulse rate after walking(1) • repeat for jogging and running(1) • rest between each exercise(1)		4
(c)	exercise for the same amount of time/same terrain/environmental conditions/wear same clothing		1
(d)	breathing/metabolism/respiration/blood pressure/sweating		1

Total for Question 2 = 9 marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 5 of 11 · Question 5 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number				
5 (a)	(i)	• axes labelled (1) • suitable scale (1) • independent variable on X axis (1) • correct plots (2) • suitable curve (1)	-1 for each incorrect plot	6
	(ii)	• mean/average calculated (1) • reliable / reproducible / repeatable (1) • eliminate anomalies (1)		2
(b)	(i)	70 x 100 (1) 90 78%/77.78/77.8 (1)	correct answer = full marks	2
	(ii)	• travels in the blood (1) • time to reach/affect nerve (1)	R heart	2
(c)		• carries impulses to heart muscle (1) • reduces rate of contraction (1)		2

Total for question = 14 marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 5 of 11 · Question 5 part c · 2 marks

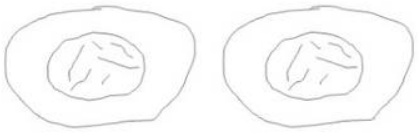
2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 6 of 11 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a) (i)	156 – 72; 84;	ECF if candidates use incorrect values from the graph with correct answer from for incorrect values = 1 mark	2
(ii)	**C** 9 minutes;		1
(iii)	In the following order only: • oxygen; • diffusion; • aerobic/cellular; • decreases/reduces/goes down;		4
(b) (i)	 must have 6 chromosomes shape not important	One mark for each correct cell drawn	2
(ii)	**B** metaphase; A is incorrect as chromosomes do not align at the equator during anaphase C is incorrect as chromosomes do not align at the equator during prophase D is incorrect as chromosomes do not align at the equator during telophase		1
			Total 10

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 6 of 11 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

uestion number	Answer	Notes	Mark s

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 7 of 11 · Question 4 part b · 4 marks

2023 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

<table>
<tr>
<td>4 (a) (i)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate
in beats per minute</th></tr>
<tr><td>walking</td><td></td></tr>
<tr><td>jogging</td><td></td></tr>
<tr><td>running</td><td></td></tr>
</table>
</td>
<td>1 mark for valid table
1 mark for labels;
1 mark for units;</td>
<td>3</td>
</tr>
<tr>
<td>(ii)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate
in beats per minute</th></tr>
<tr><td>walking</td><td>94</td></tr>
<tr><td>jogging</td><td>116</td></tr>
<tr><td>running</td><td>133.5</td></tr>
</table>
</td>
<td>2 marks for all readings correct.
1 mark for 1 or 2 readings correct.

tolerance +/- ½ square</td>
<td>2</td>
</tr>
<tr>
<td>(iii)</td>
<td>Any four from five

• take resting pulse;
• method for measuring pulse rate e.g. finger on wrist/neck/electronic device;
• carry out an activity/exercise (for a set period of time/given time);
• measure pulse rate after activity;
• repeat (for each activity);</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>(b)</td>
<td>Any four from

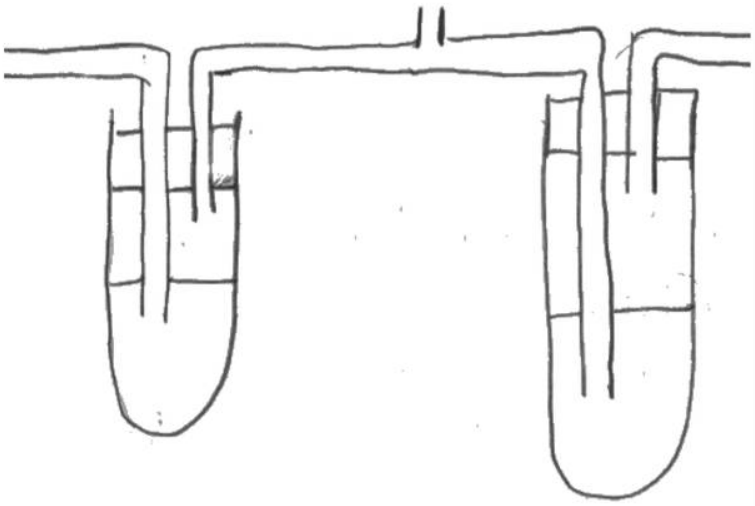
• faster/more/greater muscle contraction during running;
• more energy/ATP required;
• faster rate of respiration;
• more oxygen/ glucose required/more CO_2 removed;
• increase blood flow/faster blood flow/more blood (to muscles);</td>
<td>ORA for all marking points</td>
<td>4</td>
</tr>
</table>

Total for Question 4 = 13 marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 8 of 11 · Question 4 part b.ii · 4 marks

2024 JUN · Paper 01 · find part b.ii in this mark scheme extract

4	(a)	(i)	• apparatus correctly set up; • long tube in indicator; • short tube above indicator; 		(3)
		(ii)	Inhale and exhale gently/sterilise/clean mouthpiece/don't inhale liquid/care in assembling apparatus to avoid breaking glass;		(1)
	(b)	(i)	exercise/rest for the same length of time/at the same pace for each repeat/same student;		(1)
		(ii)	any four from • breathing rate increases (with vigorous exercise); • more oxygen inhaled/supplied; • for aerobic respiration; • more energy released/more ATP; • breakdown of lactic acid; • produced by anaerobic respiration;	ignore ref CO_2	(4)

Total for Question 4 = 9 marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 9 of 11 · Question 2 part b · 1 mark

2022 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Number				
2	(a) (i)	an answer that makes reference to three of the following: • warms air; • moistens air; • layer of (sticky) mucus; • removes/traps, pathogens/bacteria/microorganisms/dust / dirt (from entering lungs); • reference to 1st line of defence;	Mp3 must be linked to mp4 for mark	3
	(ii)	• prevents collapse; • to allow air to flow freely (to lungs/alveoli); • allows neck to bend;		2
	(iii)	B; (bronchi)		1
	(b)	D; (medulla oblongata)		1
	(c)	$17 \times 0.50 = 8.5\ dm^3$; 4×8.5; $= 0.34\ dm^3$;		3
	(d)	• reduced oxygen uptake by haemoglobin/forms carboxyhaemoglobin; • less oxygen to developing fetus; • lower rate of respiration/less energy released (by fetus); • less growth/lower birth mass;		4

Total 14 marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 9 of 11 · Question 2 part b · 1 mark

2022 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 10 of 11 · Question 2 part b.iv · 4 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.iv in this mark scheme extract

Number					
2	(a)	(i)	nose/nasal cavity/mouth, trachea/windpipe, bronchi, bronchioles;;	one mark for two correct two marks for four correct	2
		(ii)	• large surface area; • thin epithelium/walls;	R cell wall/cell membrane	2
	(b)	(i)	B; 0.04% A, C and D are all incorrect figures		1
		(ii)	• 1200×16; • $\frac{19200}{1000} = 19\ dm^3$;	ECF if any value used from table Accept 19.2	2
		(iii)	• $\frac{20 \times 10\,500}{100}$; • $\frac{2100 \times 20}{100}$; • $= 420\ cm^3$;	2 mark ECF	3
		(iv)	• more carbon dioxide in blood/lungs/body; • detected by (chemo)receptors (in arteries); • rate of breathing/tidal volume increases; • increases to remove carbon dioxide;		4

Total 14 marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 10 of 11 · Question 2 part b.iv · 4 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain heart and breathing changes in exercise and how carbon dioxide controls breathing

Answer 11 of 11 · Question 7 part b · 2 marks

2024 JUN · Paper 01 · find part b in this mark scheme extract

7 (a)		• intercostal muscles contract; • ribs move up and out; • diaphragm contracts and flattens; • air is drawn into the lungs;	ignore ref to internal/external	(4)
(b)		• detect pH of blood/levels of CO_2; • to maintain homeostasis/regulate breathing/heart rate;		(2)
(c)	(i)	tidal (volume);		(1)
	(ii)	residual (volume);		(1)
	(iii)	vital capacity;		(1)

Total for Question 7 = 9 Marks